

CCAF: Coarse-to-fine Cross-Modal Alignment and Fusion for Multimodal Sentiment Analysis

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Abstract

Multimodal sentiment analysis (MSA) has witnessed remarkable advancements in recent years. Existing MSA methods focus primarily on learning coarse-grained representations from different modalities to perform global cross-modal alignment or fusion. However, these approaches often neglect fine-grained valuable sentimental clues derived from local cross-modal interactions. Furthermore, the cross-modal alignment and fusion of complex global and local cross-modal information pose significant challenges in MSA tasks. To address this issue, we propose a novel MSA framework that simultaneously captures coarse-grained and fine-grained cross-modal sentiment cues through global and local cross-modal alignment and fusion. Our approach consists of three key components: i) optimal transport-based global and local cross-modal alignment, which separately aligns valuable global and local sentiment clues across modalities, ii) global and local cross-modal gated attention, which respectively fuse the aligned global and local cross-modal representations, and iii) prototype-informed information bottleneck, which utilizes learnable sentiment prototypes and contrastive prototype match to eliminate redundant cross-modal information at both global and local levels. Extensive experiments conducted on two publicly available MSA datasets demonstrate the effectiveness and superiority of our proposed model.

CCS Concepts

• Information systems → Data mining.

Keywords

Multimodal Sentiment Analysis, Information Bottleneck, Prototype Learning

ACM Reference Format:

Xianbing Zhao, Shengzun Yang, and Buzhou Tang. 2026. CCAF: Coarse-to-fine Cross-Modal Alignment and Fusion for Multimodal Sentiment Analysis.

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© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792569>

In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 10 pages.
<https://doi.org/10.1145/3774904.3792569>

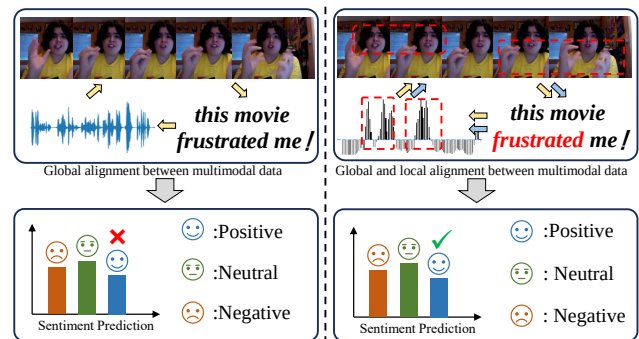


Figure 1: Previous works have primarily focused on global alignment and fusion across multimodal sequences. In contrast, our proposed approach not only considers global cross-modal correlations, but also explicitly models local dependencies, such as those among consecutive video or audio frames, to capture fine-grained local cross-modal correlations.

1 Introduction

The explosive growth and widespread popularity of short-form videos on social media platforms have driven significant interest from both academia and industry toward developing MSA methods [29, 32, 39, 47] to automatically identify the sentiment polarity of videos. Accurately understanding multimodal information, such as text, video, and audio, is crucial for enhancing machine comprehension capabilities. MSA is a fundamental and important task within the field of affective computing. However, it remains challenging due to the necessity for precise cross-modal alignment [41, 52, 54] and fusion [38, 39, 56] from a global perspective. Specifically, the former employs global cross-modal alignment techniques to learn cross-modal sentiment cues, while the latter leverages global cross-modal fusion methods to integrate multimodal representations.

These methods, which are based on Transformer architectures, focus exclusively on global relationships between sequences and overlook the rich sentiment cues present in local tokens, video frames, and audio frames. As illustrated in Figure 1, certain tokens in the text modality exhibit cross-modal associations with contiguous segments of frames in the visual modality within the context of MSA. These local sentiment cues should be effectively aligned and fused.

Although significant efforts have been made in recent years to address the above challenges through modeling cross-modal alignment and fusion methods [15, 32, 45, 55], these approaches mainly focus on the global interactions between two sequences. For example, they often consider all tokens in the text modality, all video frames in the visual modality, and all audio frames in the acoustic modality during cross-modal alignment and fusion. Such methods neglect the associations between local representations across modalities. Local representations contain rich sentiment cues. For instance, a contiguous sequence of video frames capturing facial expression changes in the visual modality, or a segment of audio frames reflecting energy fluctuations in the acoustic modality, both of which can provide valuable evidence for MSA models. Therefore, how to explicitly consider both global and local representations during cross-modal alignment and fusion remains an open research question.

To tackle these downsides, we propose a novel coarse-to-fine cross-modal alignment and fusion network that explicitly considers both global and local multimodal representations during cross-modal alignment and fusion. As shown in Figure 1, we first design global and local encoders to learn both the global and local representations for each modality. Next, we develop an optimal transport-based global-local cross-modal alignment module, which aligns not only the holistic representations of two sequences but also the representations of contiguous local tokens (or frames). We then introduce a cross-modal Gated Attention module to separately fuse the aligned global and local multimodal representations, further enabling the dynamic filtering of irrelevant information during cross-modal fusion. Finally, we design a prototype-informed information bottleneck, which utilizes learnable sentiment prototypes and contrastive prototype match to eliminate redundant information that may arise when fusing global and local multimodal representations. Extensive experiments conducted on two publicly available datasets demonstrate the effectiveness and superiority of our proposed model.

The main contributions of this work are threefold:

- We introduce an optimal transport-based global-local cross-modal alignment module that effectively aligns global and local cross-modal sentiment cues to facilitate efficient cross-modal fusion.
- We design a global-local cross-modal gated attention module that dynamically fuses and filters the aligned global and local multimodal representations.
- To eliminate the redundancy introduced by global and local multimodal representations, we propose a prototype-informed information bottleneck, which utilizes learnable sentiment prototypes and contrastive prototype match to

constrain the model to retain only the multimodal representations relevant to sentiment semantics.

2 Related Work

2.1 Multimodal Sentiment Analysis

MSA aims to capture sentiment information from multimodal data, such as text, audio, and video, and leverage this information to predict sentiment categories. Existing methods can be broadly classified into two categories: modality fusion-based approaches and modality alignment-based approaches.

The former integrates features from different modalities through fine-grained interactions across multiple dimensions, constructing a shared representation space for otherwise independent features and aggregating semantic information from each modality [9, 23, 32, 47, 52]. MFM [33] learns decomposed subspaces to capture both modality-shared and modality-specific representations. MISA [10] employs similarity and dissimilarity loss constraints to guide modalities in learning shared features, thereby alleviating modality heterogeneity. Self-MM [46] adopts a semi-supervised learning mechanism to refine unimodal labels. ConKi [47] leverages contrastive learning to construct modality-shared and modality-private representations. DMD [15] builds upon shared and private representations by applying knowledge distillation techniques to transfer knowledge across subspaces. Specifically, MMIM [9] proposed a hierarchical approach that integrates and fuses both cross-modal and intra-modal semantic information through a layered architecture. MFON [52] identifies modules with weaker representational capacity and enhances them through targeted prompting to improve their performance.

The latter aligns the semantic spaces of different modalities at a fine-grained level, enabling semantic-level matching and correspondence between multimodal features [16, 19, 40, 41]. Specifically, TFN [48] designs a tensor fusion framework that learns unimodal, bimodal, and trimodal interaction tensors to capture cross-modal correlations, thereby enhancing the performance of multimodal sentiment analysis models. LMF [20] addresses the high computational complexity of TFN by introducing a low-rank approximation strategy to improve computational efficiency. MuLT [32] leverages attention mechanisms to design cross-modal attention modules tailored for multimodal sentiment analysis, enabling the modeling of asynchronous temporal correlations across modalities. PMR [21] extends MuLT by proposing an iterative attention refinement mechanism that progressively enhances multimodal representations, leading to improved model performance. MAG-BERT [29] embeds multimodal features into the intermediate layers of a pre-trained language model, thereby extending its perceptual capability to multimodal data. SemiIN [18] introduces a semi-supervised learning approach that employs masked attention and gating mechanisms to model both intra-modal and inter-modal relationships. GLoMo [58] utilizes a multi-expert network to capture both global and local details, further strengthening the representational capacity of multimodal sentiment analysis models. AlignMamba [16] utilizes optimal transport to align audio and visual features with textual features, thereby enhancing the representational capacity of the audio and visual modalities. QaP [17] embeds multimodal queries as prompts into a large language model (LLM), enhancing

the model’s multimodal generation capability. MSA-Adapter [45] inserts a lightweight adapter into the language model, reducing the number of trainable parameters while improving predictive performance.

2.2 Information Bottleneck

The information bottleneck (IB) method is typically used to obtain the minimal representation that preserves all relevant information. Since the introduction of variational information bottleneck (VIB), IB-based methods have been widely adopted in the field of deep learning [1, 13, 37, 53, 57]. Specifically, Wang et al. [36] proposed a multi-view information bottleneck approach that applies the IB principle for denoising under each individual view. Zhu et al. [57] introduced a multimodal variational encoder-decoder (MMVED) framework for micro-video popularity prediction. By leveraging a variational information bottleneck and a Bayesian product-of-experts fusion strategy, their method effectively models uncertainty and reduces noise across modalities. Lee et al. [13] proposed a deep variational information bottleneck framework for incomplete multi-view omics data integration. Their approach jointly models intra-view and inter-view dependencies while effectively handling diverse view-missing patterns, achieving superior performance on real-world omics datasets. Mai et al. [24] incorporated the information bottleneck method at both the unimodal and multimodal fusion stages, and designed several IB-based strategies to enhance representation learning. Zhang et al. [53] integrated prototype learning with the information bottleneck approach and employed contrastive learning to reduce the distance between latent variables and prototypes.

Although significant progress has been achieved by incorporating either global or local information, these methods overlook the alignment of global and local information prior to cross-modal fusion, as well as the removal of redundant information that arises during the integration of global and local features. Inspired by optimal transport and information bottleneck methods [25, 41, 53], we design the MSA model that can fuse aligned global and local cross-modal features while eliminating redundancy remains an important challenge.

3 Method

In this section, we present detailed descriptions of each component of the model, as illustrated in Figure 2. Concretely, we first introduce the global and local feature extraction process. Then, we employ an optimal transport-based approach to align the global and local features across the three modalities. Afterwards, we employ a gated attention model to enable mutual learning between global and local features, allowing them to complement each other. Finally, to mitigate information redundancy during the fusion process, we employ a prototype-informed information bottleneck strategy to denoise redundant features.

3.1 Features Extraction

Given the i -th video clip, we first use the pre-trained models to extract text [28], visual [5], and acoustic [12] features, denoted as $x_{t,v,a}^i \in R^{l \times d_{t,v,a}}$, respectively, where l denotes the sequence length and $d_{t,v,a}$ denotes the feature dimensionality of the three modalities.

Then we utilize fully connected layers and convolutional layers to extract global features $x_{t,v,a}^g \in R^{l \times d}$ and local features $x_{t,v,a}^l \in R^{l \times d}$, respectively. Here d denotes the unified feature dimensionality after projection.

3.2 Optimal Transport-based Global-local Cross-modal Alignment

Global features capture long-range temporal context, enabling the learning of high-level semantics. Local features capture short-term dynamics, supporting the learning of low-level semantics. Aligning the distributions of global and local features facilitates efficient cross-modal fusion. We adopt an optimal transport (OT) approach to separately align high-level and low-level semantics. Previous studies [41] have shown that feature alignment methods based on optimal transport theory exhibit superior performance. Optimal transport theory provides a principled and flexible framework for measuring the distance between two feature distributions from heterogeneous modalities, enabling the model to learn robust alignment by minimizing the overall transport cost between them. We leverage optimal transport to compute alignment losses for both global and local features between each pair of modalities, including text, visual, and acoustic, thereby ensuring semantic consistency across modalities at multiple levels of granularity.

For notational convenience, we illustrate the alignment process using two arbitrary modalities, denoted as modality m_α and modality m_β , where $m_\alpha, m_\beta \in \mathcal{M} = \{t, a, v\}$, and define their corresponding feature representations as $E_\alpha \in R^{l_\alpha \times d}$ and $E_\beta \in R^{l_\beta \times d}$, respectively. For any token e_α^i from modality m_α and token e_β^j from modality m_β , we measure the distance between them using the Euclidean norm, which serves as the cost of optimal transport between the two tokens.

$$d(e_\alpha^i, e_\beta^j) = \frac{1}{2} \|e_\alpha^i - e_\beta^j\|_2^2, \quad (1)$$

we thus obtain the optimal transport cost matrix $C \in R^{l_\alpha \times l_\beta}$, where $C(i, j) = d(e_\alpha^i, e_\beta^j)$ represents the optimal transport distance between token e_α^i and e_β^j . The Wasserstein Distance between the two modalities can be defined as:

$$W_{ot}(m_\alpha, m_\beta) = \min_{T \in \Delta} \sum_{i=1}^{l_\alpha} \sum_{j=1}^{l_\beta} T_{ij} C(i, j), \quad (2)$$

where T represents the transport matrix, which establishes a token-level fine-grained alignment scheme between the two modalities, allowing the model to capture subtle cross-modal semantic correspondences, $T_{i,j}$ denotes the optimal transport weight between token e_α^i and e_β^j , and Δ represents the normalization constraints that the optimal transport matrix T must satisfy. By solving the optimal transport optimization problem, we obtain the optimal transport matrix T^* along with the Wasserstein Distance. We adopt this Wasserstein Distance as the alignment loss, referred to as the optimal transport loss.

$$\mathcal{L}(m_\alpha, m_\beta) = \sum_{i=1}^{l_\alpha} \sum_{j=1}^{l_\beta} T_{ij}^* C(i, j), \quad (3)$$

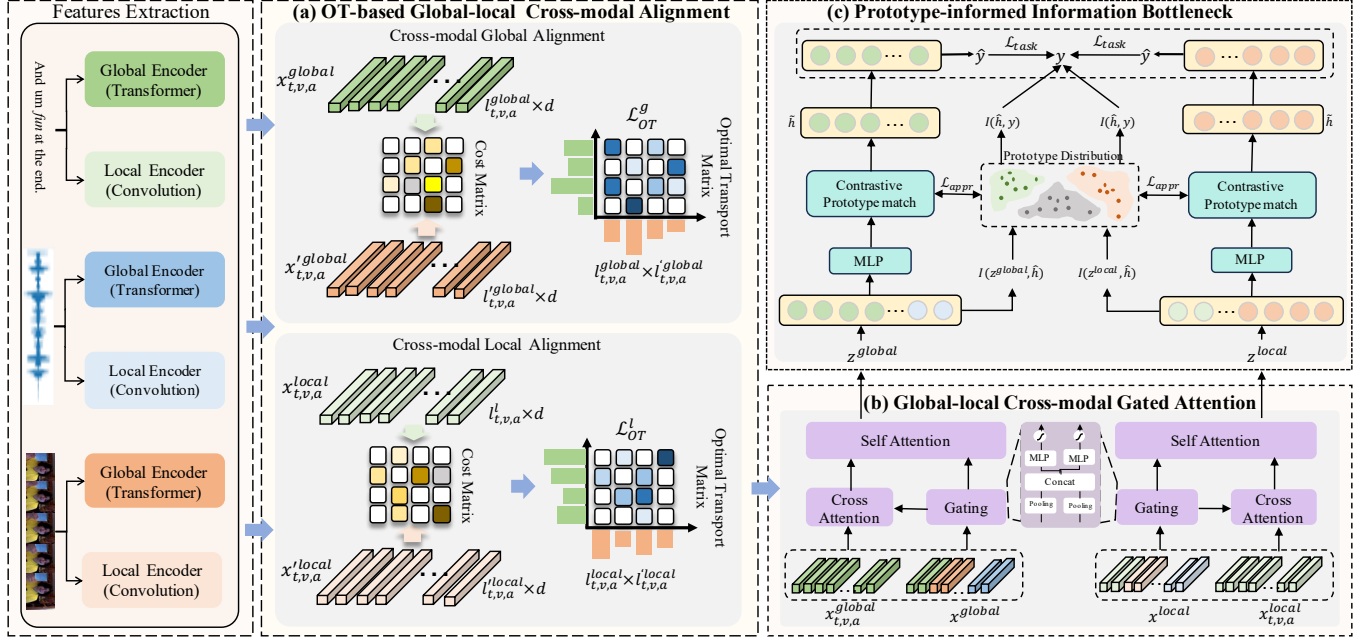


Figure 2: Schematic illustration of the proposed CCAF framework with three components. The optimal transport-based global and local alignment module simultaneously aligns both the global cross-modal sequence and fine-grained local cross-modal correlations. The prototype-informed information bottleneck is designed to eliminate redundant information related to sentiment that may arise during global and local cross-modal fusion.

where T^* represents the optimal transport matrix obtained by solving the optimization problem. To achieve fine-grained alignment, we perform alignment across the three modalities at both global and local levels. The final optimal transport alignment loss is defined as follows:

$$\mathcal{L}_{ot}^n = \sum_{i \in \mathcal{M}} \sum_{j \in \mathcal{M}} \mathcal{L}(x_i^n, x_j^n), \quad (4)$$

$$\mathcal{L}_{ot} = \lambda_1 \mathcal{L}_{ot}^g + \lambda_2 \mathcal{L}_{ot}^l, \quad (5)$$

where $n \in \mathcal{S} = \{g, l\}$ and $i \neq j$.

3.3 Global-local Cross-modal Gated Attention

To effectively integrate the holistic semantic information captured by the global features with the fine-grained sentiment cues encoded in the local features, we design a global-local cross-modal gated attention module that dynamically adjusts the degree of gated attention based on the correlation between the features [34]. The gated attention module is defined as follows:

$$z_m^n = \text{GatedAttention}(x_m^n, x^n), \quad (6)$$

where $n \in \mathcal{S}, m \in \mathcal{M}$. $x^n \in R^{3l \times d}$ represents the common message obtained by concatenating features from different modalities along the sequence dimension.

The gated attention mechanism consists of a cross-modal interaction phase and a self-attention phase. In the cross-modal interaction phase, x_m^n and x^n are sequentially processed by a cross-modal attention module with a residual connection and a feed-forward network

(FFN) module with a residual connection. In both modules, gating scores are used to adaptively regulate the proportion of information learned from the common message.

$$o = g_{mhca} \cdot \text{MHCA}(\text{LN}(x_m^n), \text{LN}(x^n)) + x_m^n, \quad (7)$$

$$v = g_{ffn} \cdot \text{FFN}(o) + o, \quad (8)$$

where $\text{LN}(\cdot)$ refers to the layer normalization operation, g_{mhca} and g_{ffn} represent gating scores predicted by the gating function. Specifically, we first concatenate the pooled representations of x_m^n and x^n as u , which is then passed through two parallel MLP layers with corresponding activation functions to produce two gating scores.

$$g_{mhca}, g_{ffn} = \sigma(\text{MLP}_{mhca}(u), \text{MLP}_{ffn}(u)), \quad (9)$$

where $\sigma(\cdot)$ represents the activation function, MLP_{mhca} , MLP_{ffn} are used to obtain g_{mhca}, g_{ffn} , respectively.

To better capture the fused semantic information, the output of the cross-modal interaction module is further processed by a self-attention module with residual connections and a feed-forward network during the self-attention phase.

$$q = \text{MHSA}(v) + v, \quad (10)$$

$$z = \text{FFN}(q) + q, \quad (11)$$

We then concatenate the enhanced global features and local features from the three modalities separately to obtain the overall global representation $z^g \in R^{l \times 3d}$ and overall local representation $z^l \in R^{l \times 3d}$.

3.4 Prototype-informed Information Bottleneck

Leveraging both global and local features significantly enriches the sentiment information available for modeling. However, it also introduces considerable redundancy, we introduce an information bottleneck mechanism, which utilizes learnable sentiment prototypes and contrastive prototype match. The goal of ontrastive prototype match leverages sentiment prototypes to match and filter multimodal features for sentiment polarity prediction. The information bottleneck mechanism is grounded in mutual information [4, 53], aiming to encourage the model to retain only the most relevant information for the target task through an intermediate encoded representation h . Inspired by these works, we use prototype-based mutual information to remove redundant information introduced during global and local multimodal fusion. The objective function is defined as follows:

$$\mathcal{D}_{ib} = I(y; h) - \beta I(z; h), \quad (12)$$

To facilitate gradient propagation, the objective function is typically optimized by minimizing the negative variational lower bound [1] :

$$J_{ib} = E_{h \sim p(h|z)} [-\log q(y | h) + \beta \cdot KL(p(h | z) \| q(h))], \quad (13)$$

Here, h is constructed from the mean and variance derived from z through a function f^θ . Our goal is to preserve detailed information across the entire set of feature vectors, whereas traditional information bottleneck methods typically model the distribution of each feature vector individually, making it difficult to capture the overall structure. To address this limitation, we propose a prototype-informed information bottleneck, which approximates the distribution of high-dimensional features $p(h | z)$ using a parameter distribution H_C composed of a set of prototype distributions, denoted as $H_C = \{p(\hat{h} | y)\}_{y=1}^C$, where C denotes the total number of categories. The parameters of each prototype distribution are initialized using Kaiming initialization[11] to ensure stable variance propagation and better convergence during training. Each prototype distribution encapsulates the sentiment information associated with its corresponding prototype, denoted as $p(\hat{h} | y) = \mathcal{N}(\hat{h}; \mu_y, \Sigma_y)$. Specifically, we establish a mapping between the input high-dimensional feature distribution $p(h | z)$ and its corresponding prototype distribution, and then utilize the prototype distribution $p(\hat{h} | y)$ to replace the original feature distribution for denoising within the information bottleneck. Therefore, the variational upper bound in the Eq.13 can be expressed as:

$$J_{ib} = E_{\hat{h} \sim p(\hat{h}|z)} [-\log q(y | \hat{h}) + \beta \cdot KL(p(\hat{h} | y) \| q(h))], \quad (14)$$

To align the overall feature distribution h with its corresponding prototype distribution $\hat{h}$, we maximize the similarity between the target prototype distribution $\hat{h}^+$ (positive prototype) and the overall feature distribution, while simultaneously minimizing the similarity between the feature distribution and other prototype distributions $\hat{h}^-$. During training, the positive prototype is determined by the labels, while during testing, it is selected as the prototype distribution most similar to h . To reduce irrelevant information, we select the *topK* tokens from the l candidates that are most similar to the positive prototype distribution $\hat{h}^+$ for alignment. Given the i -th prototype distribution $\hat{h}_i$ and the selected feature distribution

$\tilde{h}$, The similarity is defined as follows:

$$Sim(\hat{h}_i, h) = \frac{1}{K} \sum_{k=1}^K d(\hat{h}_i, h_k), \quad (15)$$

where $d(\cdot)$ represents the distance computation method. We use cosine similarity as the distance metric. The overall approximated loss is defined as follows:

$$\mathcal{L}_{appr} = -Sim(\hat{h}^+, \tilde{h}) + \frac{1}{C-1} \sum_{i=1}^{C-1} Sim(\hat{h}_i^-, \tilde{h}), \quad (16)$$

We observe that the first term in the upper bound J_{IB} focuses on learning the relationship between discriminative features and class labels. To better adapt to downstream tasks, we employ *MSE* loss to learn the relationship between the prototype distribution $\hat{h}$ and its corresponding label. The final information bottleneck loss is defined as follows:

$$\mathcal{L}(z) = \frac{1}{C} \sum_{i=1}^C \{\gamma_1 MSE(y_i, \hat{h}_i) + \gamma_2 KL(\mathcal{N}(p(\hat{h} | y) \| q(h))\}, \quad (17)$$

Here, γ_1 and γ_2 are hyperparameters. We apply the prototype-informed information bottleneck separately to the global and local features for denoising. The final loss is formulated as follows:

$$\mathcal{L}_{glib} = \sum_{n \in S} \mathcal{L}(z^n), \quad (18)$$

where $S = \{g, l\}$. To encourage the denoised global and local features to align with the same prototype distribution center, we share the parameterized distribution P_C during the denoising process. Finally, we concatenate the pooled global feature $\tilde{h}^g$ and local feature $\tilde{h}^l$ and use the combined representation for final classification.

3.5 Optimization Object

The overall optimization objective consists of four components: the loss for the MSA task, the optimal transport alignment loss, the approximated loss, and the information bottleneck loss. We adopt the MSE loss for the sentiment prediction task. The total loss is computed as follows:

$$\mathcal{L}_{total} = \mathcal{L}_{msa} + \mathcal{L}_{glib} + \mathcal{L}_{appr} + \mathcal{L}_{ot} \quad (19)$$

4 EXPERIMENTS

4.1 Datasets

CMU-MOSI[49] consists of 2,199 YouTube video segments. Each sample is manually annotated with a sentiment intensity label ranging from -3 (strongly negative sentiment) to +3 (strongly positive sentiment). **CMU-MOSEI**[50] is also collected from YouTube and consists of 22,856 video segments with a broader range of video types.

4.2 Experimental Settings

4.2.1 Evaluation Protocols. Following previous work [32], we employ five evaluation metrics to compare model performance: Binary Classification Accuracy (Acc2), F1-Score, Mean Absolute Error (MAE), Pearson Correlation (Corr), and seven-class classification

Models	CMU-MOSI				CMU-MOSEI			
	Acc2 $\uparrow$	F1 $\uparrow$	Corr $\uparrow$	MAE $\downarrow$	Acc2 $\uparrow$	F1 $\uparrow$	Corr $\uparrow$	MAE $\downarrow$
TFN	-/80.8	-/80.7	0.698	0.901	-/82.5	-/82.1	0.700	0.593
LMF	-/82.5	-/82.4	0.695	0.917	-/82.0	-/82.1	0.677	0.623
Mult	81.5/84.1	80.6/83.9	0.711	0.861	-/82.5	-/82.3	0.703	0.580
MISA	80.79/82.10	80.77/82.03	0.764	0.804	82.59/84.23	82.67/83.97	0.724	0.568
Self-MM	82.54/84.77	82.68/84.91	0.795	0.712	82.68/84.96	82.95/84.93	0.767	0.529
MMIM	84.14/86.06	84.00/85.98	0.800	0.700	82.24/85.97	82.66/85.94	0.772	0.526
FDMER	-/84.6	-/84.7	0.788	0.724	-/86.1	-/85.8	0.773	0.536
AMML	-/84.9	-/84.8	0.792	0.723	-/85.3	-/85.2	0.776	0.614
ConKI	84.37/86.13	84.33/86.13	0.816	0.681	82.73/86.25	83.08/86.15	0.782	0.529
HyDiscGAN	84.1/86.7	83.7/86.3	0.782	0.749	81.9/86.3	82.1/86.2	0.761	0.533
MFON	84.84/86.89	84.75/86.86	0.797	0.725	82.70/86.32	83.13/86.29	0.780	0.528
DMD	-/86.00	-/86.00	-	-	-/86.60	-/86.60	-	-
ConFEDE	84.17/85.52	84.13/85.52	0.784	0.742	81.65/85.82	82.17/85.83	0.780	0.522
DTN	-/86.20	-/86.20	0.807	0.714	-/86.30	-/86.30	0.788	0.579
DEVA	84.40/86.29	84.48/86.30	0.730	0.787	83.26/86.13	82.93/86.21	0.541	0.769
DLF	-/85.06	-/85.04	0.781	0.731	-/85.42	-/85.27	0.764	0.536
GLoMo	84.1/86.7	83.9/86.6	0.782	0.718	83.7/86.5	84.0/86.4	0.771	0.539
CCAF	85.82/87.04	85.75/86.89	0.790	0.743	85.90/86.54	85.91/86.53	0.794	0.592

Table 1: Comparative experiments with the best-performing MSA models conducted on the CMU-MOSI and CMU-MOSEI datasets. The values to the left and right of '/' represent the mean and maximum results across five runs, respectively. The symbol '-' indicates that the corresponding result was not reported in the original literature.

Models	CMU-MOSI		CMU-MOSEI	
	Acc2 $\uparrow$	Acc7 $\uparrow$	Acc2 $\uparrow$	Acc7 $\uparrow$
GPT-4V	90.91	61.19	87.10	49.44
Claude3-V	78.79	32.84	79.93	30.73
Gemini-V	88.34	48.68	87.14	38.86
Chatgpt	89.60	44.44	84.97	40.77
LLama2-7B	67.68	26.38	77.30	16.78
LLama2-13B	81.86	31.49	81.66	24.33
Flan-T5-XXL	89.60	42.86	86.52	46.29
QwenVL-Chat	85.93	33.92	80.41	44.12
BLIP-2	88.99	43.42	86.88	45.79
Instruct-BLIP	88.68	43.28	85.98	45.68
CCAF	91.31	48.25	88.00	53.39

Table 2: Comparison experiments with state-of-the-art multimodal large language models conducted on the CMU-MOSI and CMU-MOSEI datasets.

accuracy (Acc7). To compute Acc2 and Acc7, we convert the regression outputs into categorical predictions. For all metrics, higher values indicate better performance except MAE.

4.2.2 Implementation Details. Our model is trained on an A100 GPU with the AdamW optimizer and a batch size of 32. For the CMU-MOSI dataset, the number of epochs is set to 50 with a learning rate of $5e-5$. For the CMU-MOSEI dataset, the number of epochs is set to 20 with a learning rate of $5e-6$. The hyperparameters γ_1 and

γ_2 are set to 0.01 and 0.001, while the hyperparameters λ_1 and λ_2 are both set to 0.01. The convolution kernel size is set to 3, and the hidden layer dimension is 768.

4.3 Baselines

We evaluate the performance of our model by comparing it with state-of-the-art models in the field of MSA. These baselines can be broadly categorized into two groups.

- Methods that aim to explore multimodal representation learning or alignment, namely, TFN [48], MISA [10], FDMER [42], ConKI [47], ConFEDE [43], DTN [51], DLF [35], namely, AMML [30], DMD [15], HyDiscGAN [39], MFON [52].
- Methods that focus on advanced cross-modal fusion strategies, namely, MULT [32], LMF [20], Self-MM [46], MMIM [9], DEVA [38].

Moreover, we also include several high-performing multimodal large language models from benchmark evaluations [44] as our baselines, such as Chatgpt [26], LLama2 [31], Flan-T5 [6], Qwen-VLs [3], BLIP-2 [14], InstructBLIP [7], GPT-4V [27], Claude3-V [2], and Gemini-V [8].

4.4 Performance Comparison

Table 1 presents the comparison results between our model and existing MSA models. Among all compared methods, MFON achieves the best Acc2 and F1 scores on both datasets by specifically optimizing the underperforming modalities through targeted training. Specifically, MFON employs prompting to focus information on

the underperforming modalities and introduces a distillation strategy to enhance the utilization of information within these modalities. This result demonstrates the importance of fully leveraging intra-modal semantic information and improving intra-modal information utilization for emotion recognition. Compared to MFON, our proposed model achieves relative gains of 0.15% and 0.22% in Acc2 on the two datasets, respectively. This improvement can be attributed to our model’s ability to align features from different modalities at both global and local levels while simultaneously eliminating redundant or noisy information. By jointly optimizing cross-modal global and local alignments through Optimal Transport, the model effectively captures both coarse-grained semantic consistency and fine-grained sentiment correspondence. This dual-level alignment enables complementary information from text, visual, and audio modalities to be fully leveraged rather than being dominated by any single modality. Furthermore, the incorporation of the prototype-informed information bottleneck allows the model to selectively filter out modality-specific redundancy, ensuring that only sentiment-relevant representations are preserved. As a result, our framework maximizes the utilization of multimodal information, enhancing both the discriminative power and robustness of the learned representations.

Table 2 presents the comparison between our model and MLLMs. To ensure a fair comparison with the baselines, we replace the text encoder in our model with the pretrained T5 encoder [6]. As shown in the table 2, on CMU-MOSI, GPT-4V achieves the best performance in both Acc2 and Acc7, while on CMU-MOSEI, Gemini-V and GPT-4V achieve the best results in Acc2 and Acc7, respectively, and both generally outperform the best MSA models. This highlights the strong semantic representation capability of large models. Compared to the best results, our model achieves relative gains of 0.4% and 0.5% in Acc2 on CMU-MOSI and CMU-MOSEI, respectively, and a relative gain of 0.2% in Acc7 on CMU-MOSEI, while our Acc7 score on CMU-MOSI is second only to GPT-4V. These results further demonstrate that our model can effectively enrich and preserve informative semantic representations while reducing redundant and noisy information. While MLLMs achieve significantly better performance than traditional MSA models owing to their powerful language representation capabilities enabled by large-scale parameters, they often exhibit a strong bias toward textual cues, which can limit their effectiveness in scenarios requiring fine-grained visual understanding. In contrast, our proposed CCAF model leverages a prototype-informed information bottleneck to achieve more balanced cross-modal feature integration. Compared to the best-performing MLLM results, CCAF still achieves consistent improvements across multiple metrics, demonstrating its ability to effectively mitigate the modality bias inherent in large multimodal models. This enhancement primarily stems from CCAF’s superior cross-modal alignment mechanism, which enables it to better capture the complementary relationships between visual, audio and textual modalities.

4.5 Ablation Study

To further investigate the effectiveness of our model, we conducted ablation studies as follows: 1) w/o $\mathcal{L}_{ot}^g$, removing the global cross-modal alignment loss. 2) w/o $\mathcal{L}_{ot}^l$, eliminating the local cross-modal

Models	CMU-MOSI		CMU-MOSEI	
	Acc2 $\uparrow$	F1 $\uparrow$	Acc2 $\uparrow$	F1 $\uparrow$
w/o $\mathcal{L}_{ot}^g$	85.06	84.99	85.22	85.11
w/o $\mathcal{L}_{ot}^l$	84.91	84.91	85.20	85.19
w/o $\mathcal{L}_{glib}$	86.12	86.09	85.87	85.60
w/o Gating	86.59	86.53	86.26	86.10
CCAF	87.04	86.89	86.54	86.53

Table 3: The ablation experiments on the CMU-MOSI and CMU-MOSEI datasets.

alignment loss. 3) w/o $\mathcal{L}_{glib}$, excluding the prototype-informed information bottleneck module. 4) w/o Gating, removing the gating function from the fusion module. The detailed results are presented in Table 3.

Compared to our full model, w/o $\mathcal{L}_{ot}^g$ and w/o $\mathcal{L}_{ot}^l$ leads to the most significant performance drops, confirming the critical role of cross-modal alignment in ensuring semantic consistency across modalities. The significant drops indicate that aligning features at both the global and local levels is essential for effectively capturing complementary sentiment cues and avoiding modality imbalance during fusion. Additionally, w/o $\mathcal{L}_{glib}$ causes a noticeable decrease in performance, which verifies that the fused multimodal representations inevitably contain redundant or noisy information. The bottleneck effectively filters such irrelevant information, allowing the model to retain only sentiment-relevant representations. Finally, w/o Gating also leads to a moderate decline. This suggests that dynamically regulating the contribution of each modality during fusion helps the model focus on the most informative features while suppressing misleading or less relevant cues. Collectively, these results confirm that each module in our framework plays a complementary and indispensable role, jointly contributing to the overall performance improvement of CCAF.

These findings further validate the rationality and effectiveness of our model design. Each component plays a complementary role in enhancing cross-modal understanding—global and local alignment jointly ensure semantic coherence, the prototype-informed information bottleneck effectively suppresses redundancy, and the gating mechanism facilitates adaptive and robust fusion. Together, these modules form a unified framework that enables CCAF to achieve stable and superior performance across different datasets and backbone architectures.

4.6 Optimal Transport alignment and prototype distribution Visualization

To gain a deeper understanding of the process of aligning different modalities using the optimal transport method, we visualize the aligned global and local features, as shown in Figure 3. Specifically, in the global alignment module, OT operates on the holistic feature distributions of each modality, where it minimizes the transport cost between high-level semantic embeddings. This allows the model to establish coarse-grained semantic correspondences across modalities. As a result, global feature alignment typically occurs at the segment level, since global features encode overall

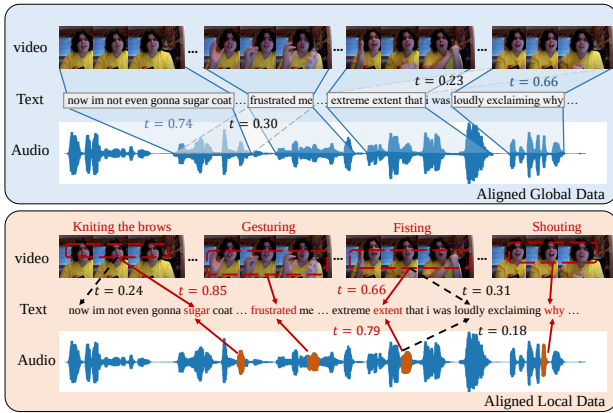


Figure 3: Visualization of global and local features aligned using the optimal transport method. For clarity, only the alignment results between the acoustic/visual modalities and the text modality are shown. Here, t denotes the weight in the optimal transport plan T .

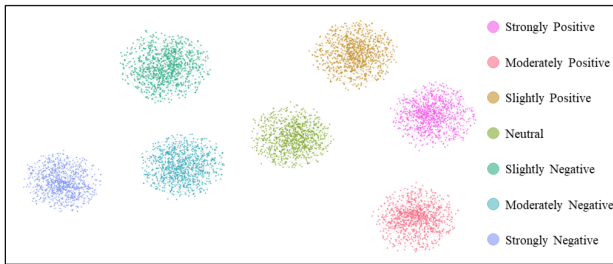


Figure 4: Visualization of prototype distributions.

sentiment trends and contextual semantics, yet lack the granularity needed to perceive subtle affective variations. In contrast, in the local alignment module, OT is applied at a finer granularity, aligning contiguous feature units (e.g., short video frame sequences or word segments). This alignment captures fine-grained sentiment cues—such as facial expression changes or voice intensity fluctuations—but is less effective in modeling long-range dependencies due to the absence of global contextual information. Consequently, the global OT alignment focuses on semantic consistency across entire modalities, whereas the local OT alignment emphasizes localized, emotion-relevant interactions. These visualized differences confirm that OT plays complementary roles at two distinct levels: global OT alignment ensures holistic semantic coherence, while local OT alignment enhances the capture of detailed emotional signals. Together, they enable the model to balance contextual understanding and sentiment sensitivity, which is essential for accurate multimodal affective analysis.

Furthermore, we also visualize the learned prototype distributions. Specifically, we sample 1,000 points from each prototype distribution and apply t-SNE [22] to reduce the dimensionality. The results are presented in Figure 4. We can observe that there is a high degree of separation between different prototype distributions,

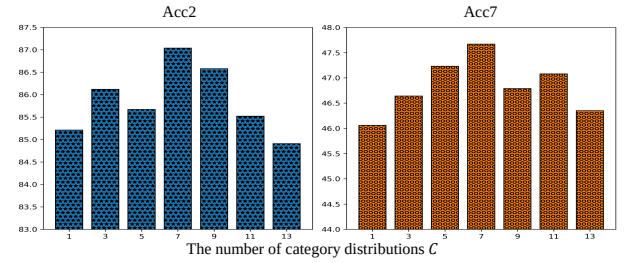


Figure 5: Analysis of the impact of the number of prototype distributions on model performance based on the CMU-MOSI dataset.

indicating that our defined distributions can effectively distinguish between different categories.

4.7 prototype Distribution Analysis

To validate the effectiveness of the prototype distributions, we conducted experiments to investigate the impact of the number of prototype distributions C on model performance. The results are shown in Figure 5. As illustrated in Figure 5, both Acc2 and Acc7 initially increase and then decrease as the number of prototype distributions increases, reaching their peak when C is set to 7.

This indicates that an appropriate number of prototype distributions is crucial for maintaining a balance between semantic granularity and representational compactness. When C is too small, each prototype covers overly broad sentiment semantics, blurring fine-grained distinctions. Conversely, an excessively large C causes overlaps among adjacent prototypes, introducing redundancy and noise. Setting $C = 7$ achieves an optimal balance, allowing the model to capture detailed yet distinct sentiment information, thus enhancing overall performance and interpretability.

5 Conclusion

We propose a coarse-to-fine cross-modal alignment and fusion framework. First, we employ global and local encoders to learn coarse-grained and fine-grained representations, respectively. We then align these global and local representations using an optimal transport-based technique. Next, we design a cross-modal gated attention mechanism to separately fuse the global and local representations. Finally, to eliminate redundancy introduced during fusion, we incorporate a prototype-informed information bottleneck to compress the global and local multimodal representations. Experiments conducted on two publicly available multimodal sentiment datasets demonstrate the effectiveness and superiority of our proposed model.

Acknowledgments

This study is partially supported by Shenzhen Science and Technology Research and Development Fund (KJZD20240903102802003), Shenzhen Science and Technology Research and Development Fund for Sustainable Development Project (GXWD20231128103819001) and Guangdong Provincial Key Laboratory Grant (2023B1212060076).

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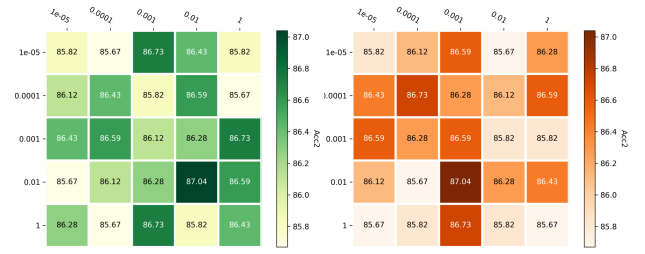


Figure 6: Analysis of the impact of hyperparameters on model performance. The left figure presents the analysis of hyperparameters λ_1 and λ_2 , while the right figure shows the analysis of hyperparameters γ_1 and γ_2 .

A Parameter Analysis

To investigate the impact of hyperparameters on model performance, we conducted a series of experiments with different hyperparameter combinations. The results are presented in Figure 6. As shown in the figure 6, our model consistently achieves stable performance across different hyperparameter settings, demonstrating strong robustness. In the left figure, the best performance is achieved when both hyperparameters λ_1 and λ_2 are set to 0.01. In the right figure, the optimal results are obtained when γ_1 is set to 0.01 and γ_2 is set to 0.001. These findings suggest that both global feature alignment and local feature alignment are equally important, further validating the necessity of incorporating local features to capture detailed sentiment information.